# Hacking_Proposal

## Who I Am / My Artistic Practice

I’m giogio. My creative practice usually begins with observing everyday experiences, bodily perception and my surrounding environment. At the moment, my work mainly focuses on two areas: **Embodied Perception** and **Human–Nonhuman Relations**.//1

My work often combines digital technologies with hands-on making. For example, I experiment with processing, assembling and combining physical materials such as paper and wood to explore their textures and possibilities.//2

Through the learning and discussions over the past half-semester, I have come to understand choreography as a way of organising relationships between bodies, movements, time and space, rather than simply arranging human movements.//3

Therefore, in my practice, I am interested not only in how the human body moves, but also in how objects, materials and machines respond to, influence and transform one another. By designing rules, interactive mechanisms and temporal processes, I organise the relationships between different elements, allowing the work to gradually unfold through their interactions. I am interested in exploring how these elements collectively form a dynamic system.//4

## Project Description

My project is a live performance that creates a continuous **Translation System**, consisting of the following stages:

**Artist's Text → Performer's Body → Computer Recognition → New Text**

The project aims to decentralise the agency of the artist and performer within a performance.//5

Starting with a story written by the artist, a performer interprets the text through their own bodily movements, without receiving any instructions or guidance. These movements are then captured and recognised by a computer system, which translates them back into a new written narrative.//6

Through this process, I aim to develop a system of translation that moves from text to body, from body to machine, and from machine back to text.//7

During this process, the original story may undergo changes, distortions and rearrangements. I want to explore how meaning shifts through different forms of translation, and how authorship and agency are redistributed between the artist, the performer and the machine.//8

## Relevance to Computational Arts

This project combines body tracking, motion recognition and rule-based text generation.

The computer acts as a mediating system, transforming bodily movements into textual information through predefined mapping rules.//9

Rather than attempting to truly understand the story, the computer follows its own recognition rules, allowing new narratives to emerge from the relationship between human bodily movement and computational processes.//10

# Tools and Technical Approach

I will use **p5.js** as the main programming environment and interactive interface to build the entire translation system. It will handle the camera feed, the visualisation of movement data, text input and recording, as well as the generation of the final text.//11

I will use **ml5.js** as the machine learning library, specifically its **BodyPose model**, to detect the performer's body posture and keypoints through a camera. This will provide the data needed for subsequent movement recognition and text translation.//12

First, I will write a short story or script containing only narrative content, without any instructions about movement, gestures, emotions or performance.

I will then recruit a performer to collaborate with me. Without receiving any additional instructions, the performer will encounter the text and interpret it through their own reading and understanding, translating each sentence into bodily movements. They will then enter or annotate their interpretation of the movements for each section of the text within the system.//13

During the live performance, the performer will face the system again and perform the entire story using the movements developed during the initial interpretation.

At this stage, the computer will recognise the performer's bodily movements in real time. Based on the previously established mappings between movements and text, the system will translate the recognised movements back into written language, gradually generating a new text.//14